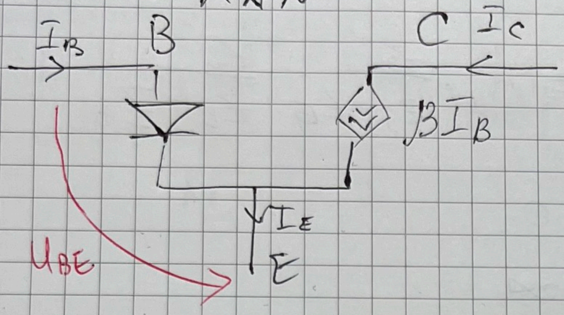
Deea
curs

Regim Dinamic

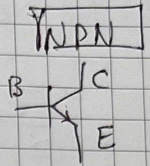
16.11.2023

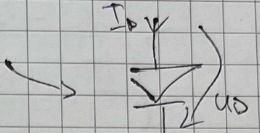
– Aprox semnal Mic –

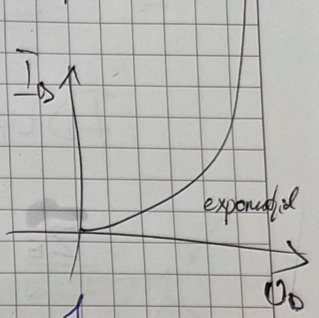
T – BIP RAN



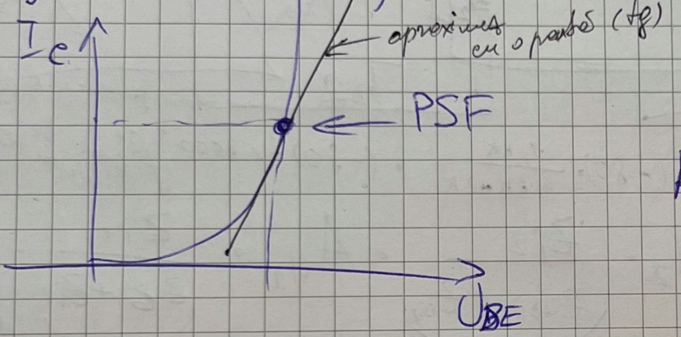
I_B
B
C
I_C
βI_B
U_{BE}
I_E
E







$$I_B \approx e^{u_D} \Rightarrow I_C = \beta I_B \approx e^{u_D}$$

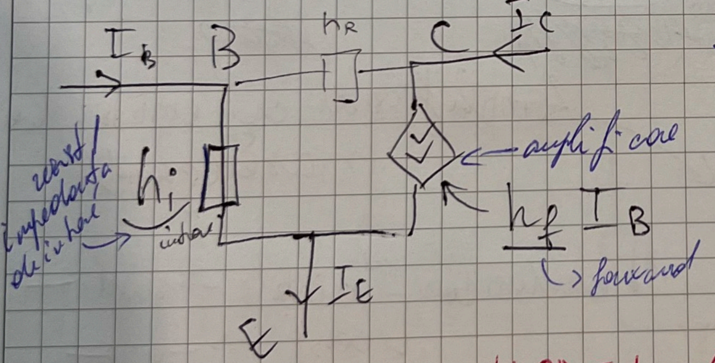


aproximarea cu o pantă (tg)

PSF

$$\underbrace{R_D}_{\text{dinamic}} = \frac{1}{40 \underbrace{I_D}_{mA}} \, K\Omega$$

$$R_{D\,TR\,BAZA-EMITOR} = \frac{1}{40 I_B} = \frac{\beta}{40 I_{C(mA)}} \, K\Omega$$



I_B
B
h_R
C
I_C
h_i
I_E
E

amplificare

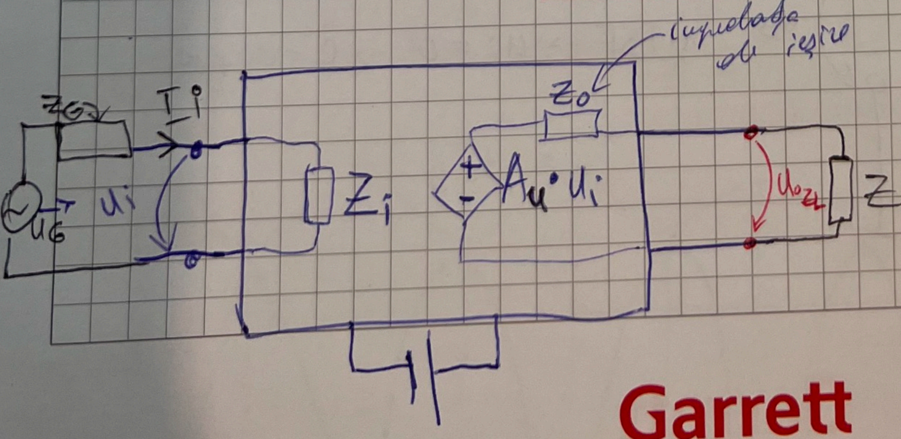
$\underbrace{h_f}_{\text{forward}} I_B$

$\underbrace{h_i}_{\text{intrare}}$ – rezistență / impedanța de intrare

$$\begin{cases} h_i = \dfrac{\beta}{40 I_C} \\ h_f = \beta \end{cases}$$

MODEL AMPLIFICATOR

// A_u – amplificare de tensiune



Z_G
I_i
U_i
U_G
Z_i
Z_o
$A_u \cdot U_i$
U_{OZL}
Z_L

impedanța de ieșire

Z_i – imp de intrare
Z_o – imp de ieșire
Z_L – impedanța de sarcină

Z_G, Z_i – divizor de ten

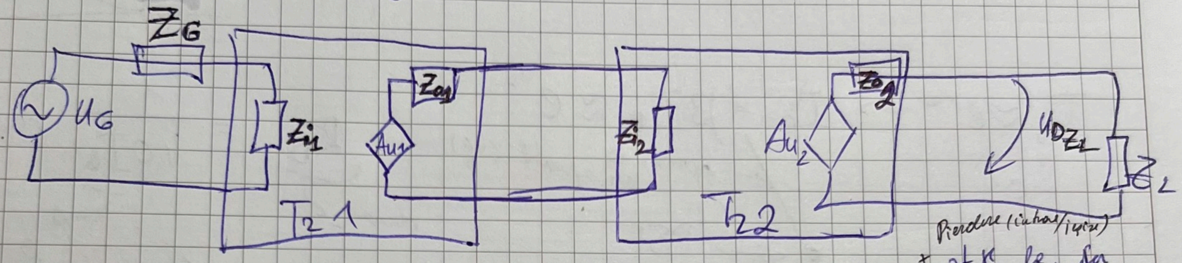
1/4

$$I_i = \frac{u_G}{Z_i + Z_G}$$

$$u_i = I_i \cdot Z_i = \frac{Z_i}{Z_i + Z_G} u_G$$

$$U_{oZ_L} = \frac{Z_L}{Z_L + Z_o} \cdot A_u \cdot u_i = A_u \cdot \frac{Z_i}{Z_i + Z_G} \cdot \frac{Z_L}{Z_L + Z_o} \cdot u_i$$

$$\text{IDEAL} \rightarrow \begin{cases} Z_o \rightarrow 0 \\ Z_i \rightarrow \infty \end{cases}$$


Z_G
u_G
Z_i1
Z_o1
A_u1
Tz 1
Z_i2
Z_o2
A_u2
Tz 2
u_oZL
Z_L

Pierdere (intrare/ieșire)

$$\text{Amplif} = (-80) \cdot \frac{1,6K}{1,6K + 2K} \cdot (-80) \sim 3000$$

cât se pierde la divizor [illegible]

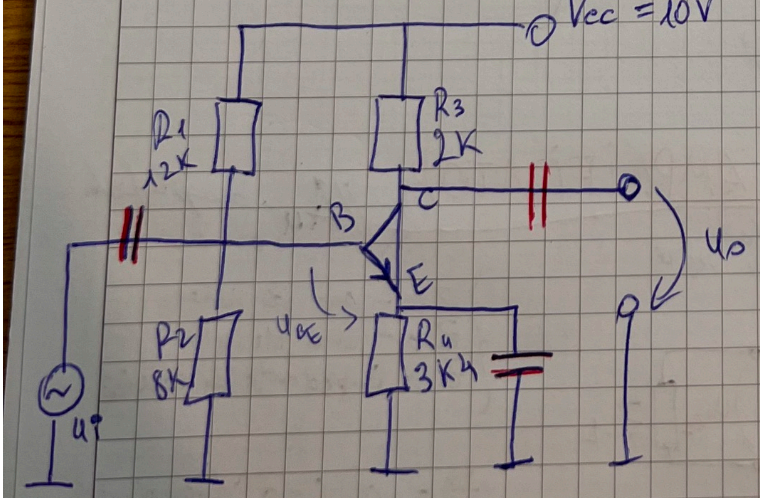
$$U_{oZ_L} = \frac{Z_L}{Z_L + Z_{o2}} A_{u2} \cdot \underbrace{\frac{Z_{i2}}{Z_{i2} + Z_{o1}}}_{\text{divizor de tensiune}} \cdot A_{u1} \cdot \underbrace{\frac{Z_{i1}}{Z_G + Z_{i1}}}_{} \cdot \underbrace{u_G}_{\text{tensiune amplificată}}$$

(~~[illegible]~~ ideal 0)

(~~[illegible]~~)

Amplificator cu emitor la masă (ground)

→ Schemă de Polarizare


Vcc = 10V
R1
12K
R3
2K
B
C
E
u_o
R2
8K
u_BE
R4
3K4
u_i

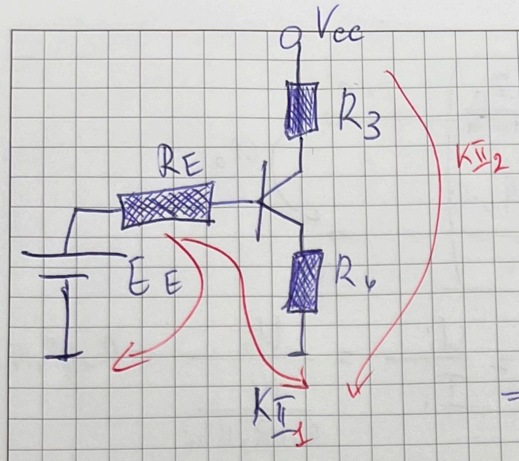
- conducătoare cu variabile → scurt —
  [const → —o o—
- Liniar → superpoziție

1) PSF → $u_i = 0$ → C - dispare

$$I_C = \beta I_B$$

Garrett
ADVANCING MOTION

de acest curs



V_{cc}
R_3
R_E
K_{II_2}
E_E
R_4
K_{II_1}

$R_E = R_1 \parallel R_2 = \frac{96}{20} = 4,8K$

$E_E = \frac{R_2}{R_1+R_2} \cdot V_{cc} = \frac{8}{8+12} \cdot 10V = 4V$

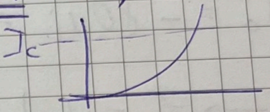
$\underline{\underline{K_{II}}}$: $E_E = R_E I_B + U_{BE} + R_4 I_E \Rightarrow$

$\Rightarrow \frac{4V - 0,6V}{\frac{4,8K}{100} + 3,4K} = 1mA$

0 ← // scap de el

- $U_{CE} = V_{cc} - I_C R_3 - I_E R_4 = \underline{\underline{4,6V}} > 0,6V$ ✓

$\Rightarrow \underline{\underline{RAN}}$



$h_i = \frac{\beta}{40 I_c} = \frac{100}{40 \cdot 1mA} = 2,5K$

$h_f = \beta = 100$

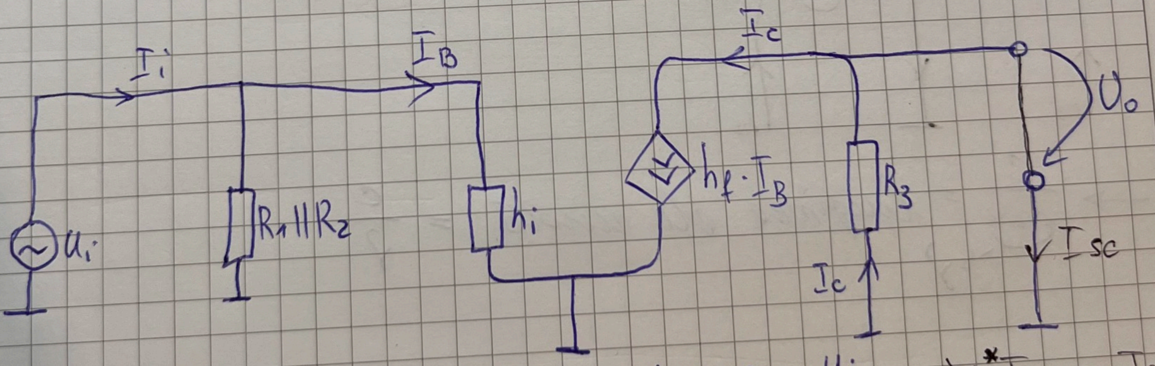
2) Regim Dinamic

$V_{cc} = 0V \rightarrow u_i$ —o—

↓ PASIVIZĂM

$u_i = I_B \cdot h_i$

$\rightarrow I_B = \frac{u_i}{h_i}$

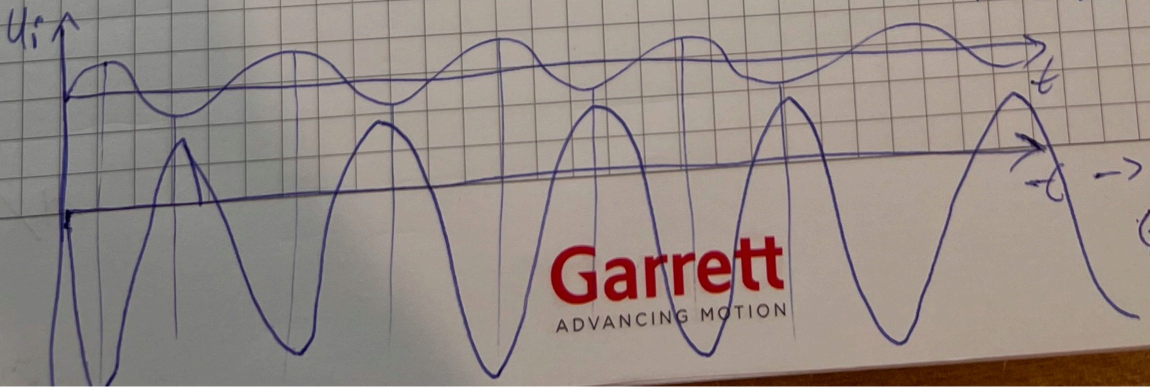


I_i
I_B
I_c
u_i
$R_1 \parallel R_2$
h_i
$h_f \cdot I_B$
R_3
U_o
I_{sc}
I_c

$\hookrightarrow {}^{*} I_{sc} = -I_c R_3$

$\underline{U_o} = -I_c \cdot R_3 = -h_f I_B R_3 = -h_f \cdot R_3 \cdot \frac{u_i}{h_i}$

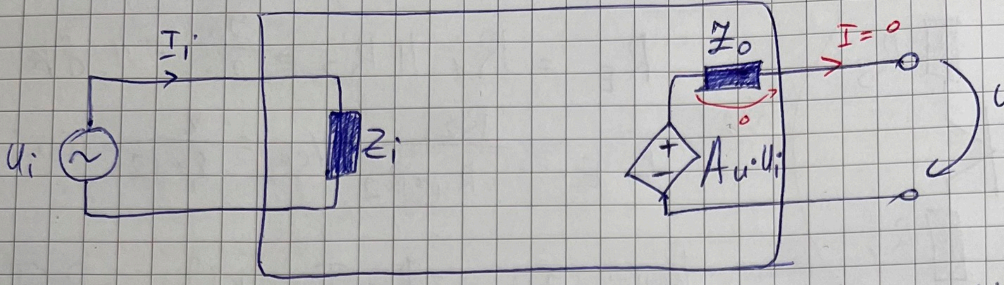
$= -100 \cdot 2k \cdot \frac{u_i}{2,5k} = \underline{-80u_i}$



→ de 80x ↗

(–") → semnalul e invers

3/4



$$I_i = \frac{u_i}{R_1 \| R_2 \| h_i} \Rightarrow Z_i = \frac{u_i}{I_i} = \frac{u_i}{\frac{u_i}{R_1 \| R_2 \| h_i}} = R_1 \| R_2 \| h_i =$$

$$= \frac{\overset{R_1 \| R_2}{4,8k} \cdot 2,5k}{4,8k + 2,5k} = \frac{12,5}{7,5} = 1,6k \Rightarrow \boxed{Z_i = 1,6k}$$

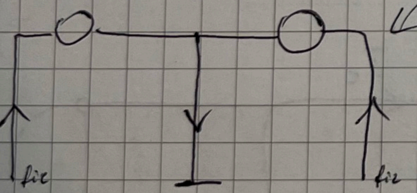
- $U_o = -80u_i = A_u \cdot u_i \big|_{I_o = 0} = \boxed{A_u = -80}$

$$\to Z_o = \frac{U_o \big|_{Z_L = \infty}}{I_{o\,\underline{sc}}} = \frac{-80u_i}{-I_c} = \frac{80\cancel{u_i}}{\frac{\cancel{u_i}}{\underset{= 2,5K}{h_i}} \cdot \beta} = \frac{2,5K \cdot 80}{100} =$$

sc → scurt circuit

$$\Rightarrow \boxed{Z_o = 2K}$$

Structură căști



$$\to \frac{c}{\underbrace{\text{frecv}}_{\nu}} = \text{lungimea de undă} = \frac{c}{\nu}$$